

# Setup - Preparation Before You Begin

## Install R, RStudio and Packages

### Steps needed to get started

1. Install R
  2. Install RStudio Desktop
  3. Install R Packages
  4. Run a quick test
- 

### 1. Install R

#### R versus RStudio

Note: **R** is the name of the programming language itself and **RStudio** is an integrated development environment (IDE) which is an enhanced interface for better organization, files management and analysis workflows.

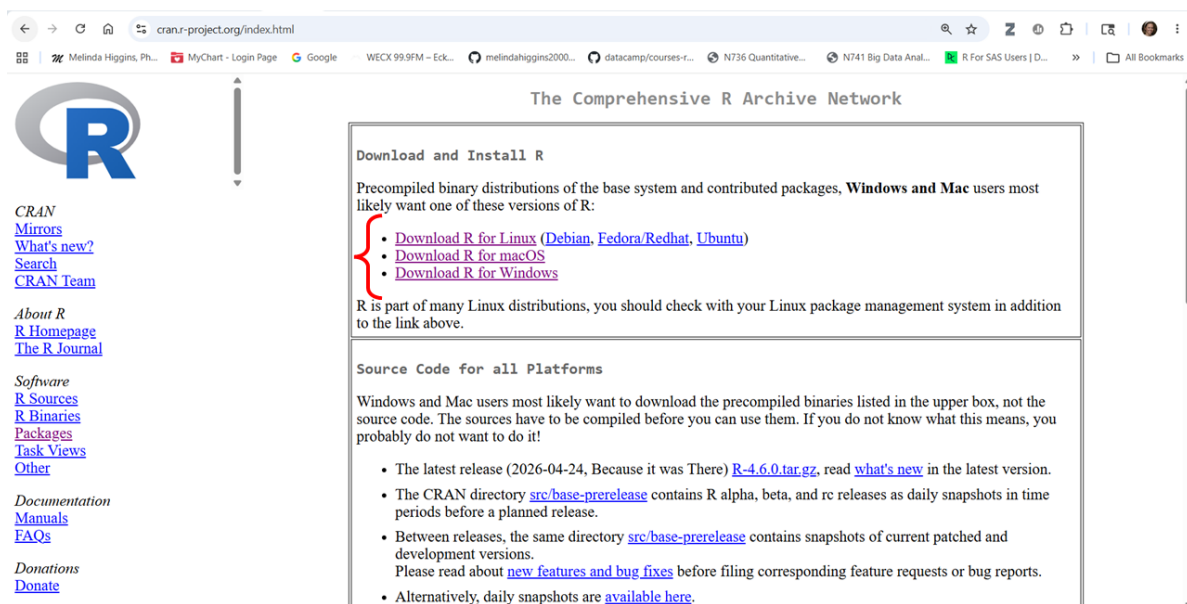
### Download R from CRAN

The R programming language is maintained and distributed by CRAN [The Comprehensive R Archive Network](#). They provide the installers for the binary distributions of the base system and contributed packages. R is FREE!

Installation Instructions are provided for these operating systems:

- [Windows](#)
  - Choose “base” and I suggest you accept the default installation options.
  - OPTIONAL - You may also want “RTools” if you think you may want to do build your own R packages in the future. This is NOT needed for this course.

- [Mac](#)
  - Please make note of the specific details on which installer you may need for your version of Mac OS (operating system).
  - I work on a Windows operating system, so please reach out to your local IT support for help getting the correct version installed for your system.
- [Linux](#)
  - Odds are you are probably using either Windows or Mac OS, but if you using Linux, please read carefully the specific details for your version of Linux.



## Other Helpful Resources For Installing R

- [Chapter 1 of Statistical Inference via Data Science: A ModernDive into R and the Tidyverse \(Second Edition\)](#) by Chester Ismay, Albert Y. Kim, and Arturo Valdivia.
- [Chapter 3.4 of The Epidemiologist R Handbook](#)

## 2. Install RStudio Desktop

### ! Install R First and RStudio Second

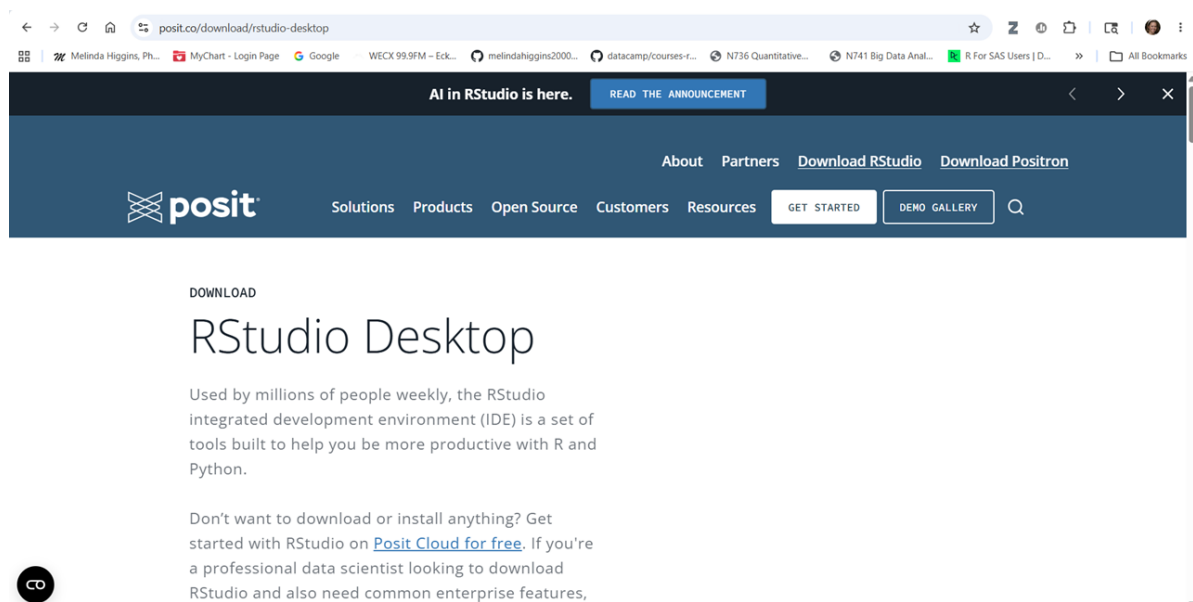
It is important to install the R programming language FIRST. I also encourage you to use the default installation options, so that when you install the RStudio Desktop application, it will “know” where (folder locations) to look for your R installation.

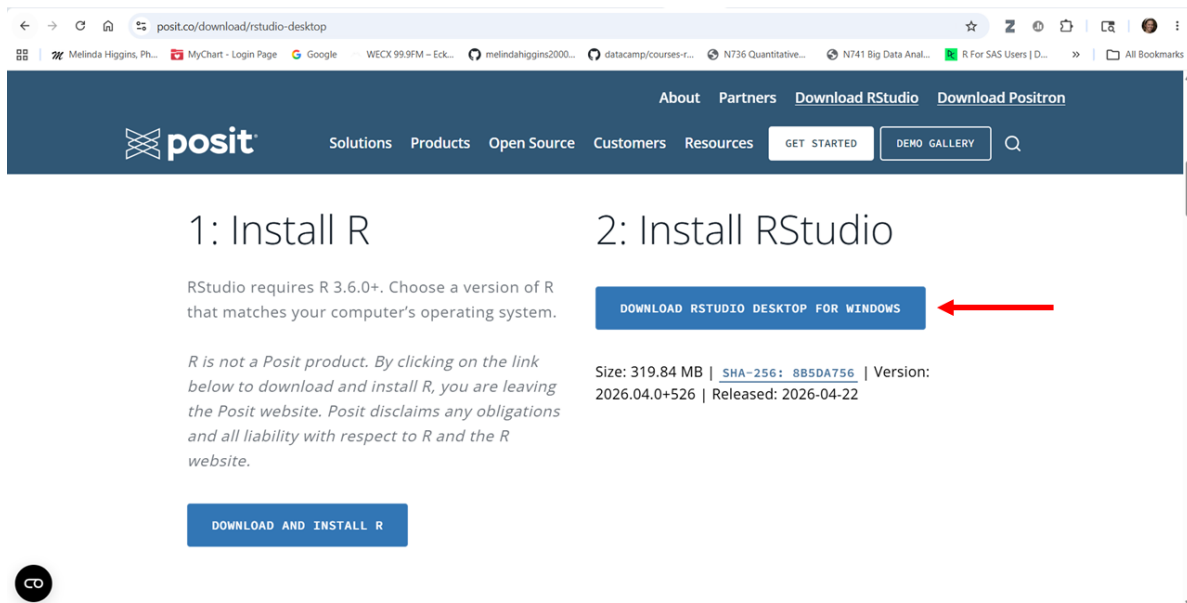
The RStudio IDE (integrated development environment) software is developed and distributed by the [POSIT Company](#), which was formerly known as “RStudio” founded in 2009. In 2022 they changed their name to “Posit”, however, the name of the software application stayed “RStudio”.

For this course, we will be using the RStudio Desktop software. RStudio Desktop is FREE. The “PRO” version does cost money.

You can download RStudio Desktop from <https://posit.co/download/rstudio-desktop/>.

I think the website link above will “autodetect” which operating system you have. Mine defaults to the Windows installer.





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## 1: Install R

RStudio requires R 3.6.0+. Choose a version of R that matches your computer's operating system.

*R is not a Posit product. By clicking on the link below to download and install R, you are leaving the Posit website. Posit disclaims any obligations and all liability with respect to R and the R website.*

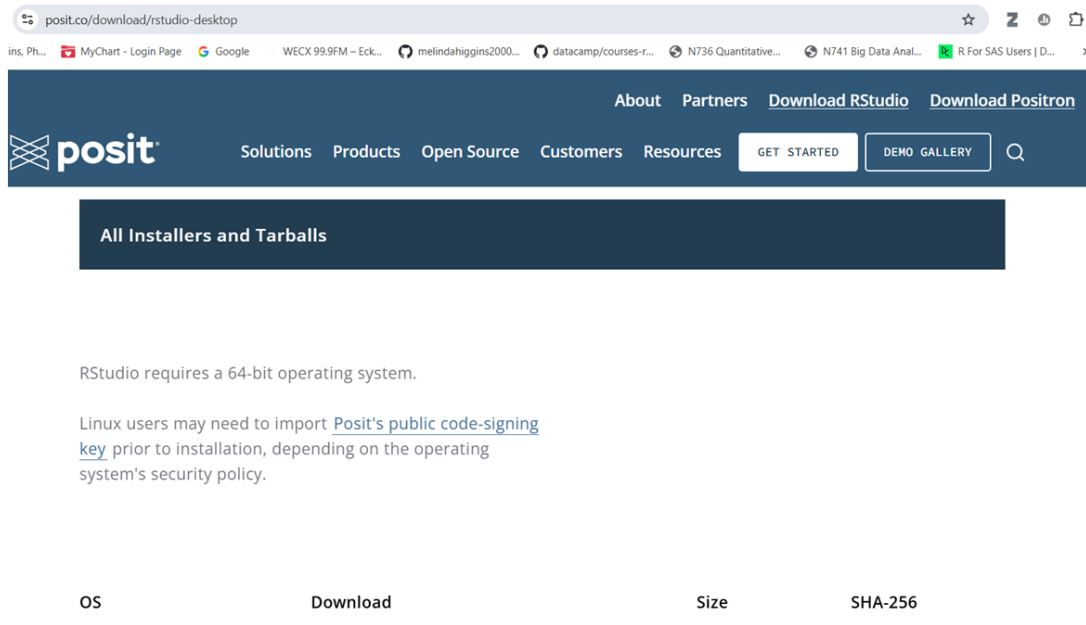
DOWNLOAD AND INSTALL R

## 2: Install RStudio

DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS

Size: 319.84 MB | SHA-256: 8B5DA756 | Version: 2026.04.0+526 | Released: 2026-04-22

However, if you have a Mac you should get the installer for the Mac OS. Regardless, you can always choose the operating system and version you need.



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### All Installers and Tarballs

RStudio requires a 64-bit operating system.

Linux users may need to import [Posit's public code-signing key](#) prior to installation, depending on the operating system's security policy.

| OS | Download | Size | SHA-256 |
|----|----------|------|---------|
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| <div><div>posit</div><div><a href="#">About</a> <a href="#">Partners</a> <a href="#">Download RStudio</a> <a href="#">Download Positron</a></div></div>         |                                                                  |           |                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------|--------------------------|
| <div><div><a href="#">Solutions</a> <a href="#">Products</a> <a href="#">Open Source</a> <a href="#">Customers</a> <a href="#">Resources</a></div><div><a href="#">GET STARTED</a> <a href="#">DEMO GALLERY</a> <input type="text"/></div></div> |                                                                  |           |                          |
| OS                                                                                                                                                                                                                                               | Download                                                         | Size      | SHA-256                  |
| <b>Windows OS</b>                                                                                                                                                                                                                                |                                                                  |           |                          |
| Windows 11                                                                                                                                                                                                                                       | <a href="#">RSTUDIO-2026.04.0-526.EXE</a> <small>⬇</small>       | 319.84 MB | <a href="#">8B5DA756</a> |
| <b>Mac OS</b>                                                                                                                                                                                                                                    |                                                                  |           |                          |
| macOS 14/15/26                                                                                                                                                                                                                                   | <a href="#">RSTUDIO-2026.04.0-526.DMG</a> <small>⬇</small>       | 694.56 MB | <a href="#">F49B0016</a> |
| <b>Linux OS Versions</b>                                                                                                                                                                                                                         |                                                                  |           |                          |
| Ubuntu 22                                                                                                                                                                                                                                        | <a href="#">RSTUDIO-2026.04.0-526-AMD64.DEB</a> <small>⬇</small> | 238.98 MB | <a href="#">AF4D87C0</a> |
| Ubuntu 24                                                                                                                                                                                                                                        | <a href="#">RSTUDIO-2026.04.0-526-AMD64.DEB</a> <small>⬇</small> | 238.98 MB | <a href="#">AF4D87C0</a> |
| Debian 12                                                                                                                                                                                                                                        | <a href="#">RSTUDIO-2026.04.0-526-AMD64.DEB</a> <small>⬇</small> | 238.98 MB | <a href="#">AF4D87C0</a> |

And if you keep scrolling down the webpage you can find older versions of RStudio if/as needed as well.

### RStudio or Positron?

For this course, I will use the RStudio IDE. However, there is a newer alternative to “RStudio”, known as “Positron” which was released in 2025.

Learn more about [Positron](#). Positron is also an IDE programming environment, but is designed to accommodate multiple programming environments and multiple sessions at the same time, e.g. you can run both R and Python and manage multiple versions at one time. We will NOT cover these details in this course.

Technically, you can also use the RStudio IDE to execute and manage Python programs as well. In fact RStudio can help manage several other programming languages as well.

### User Guides

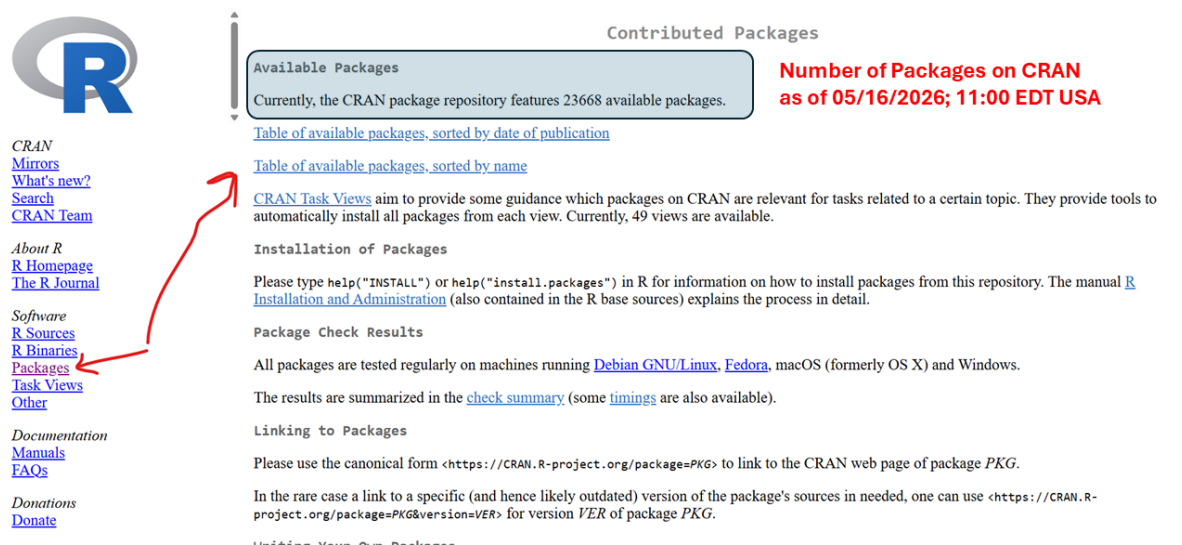
- [RStudio User Guide](#).
- [Positron User Guide](#)

### 3. Install R Packages Needed for Course

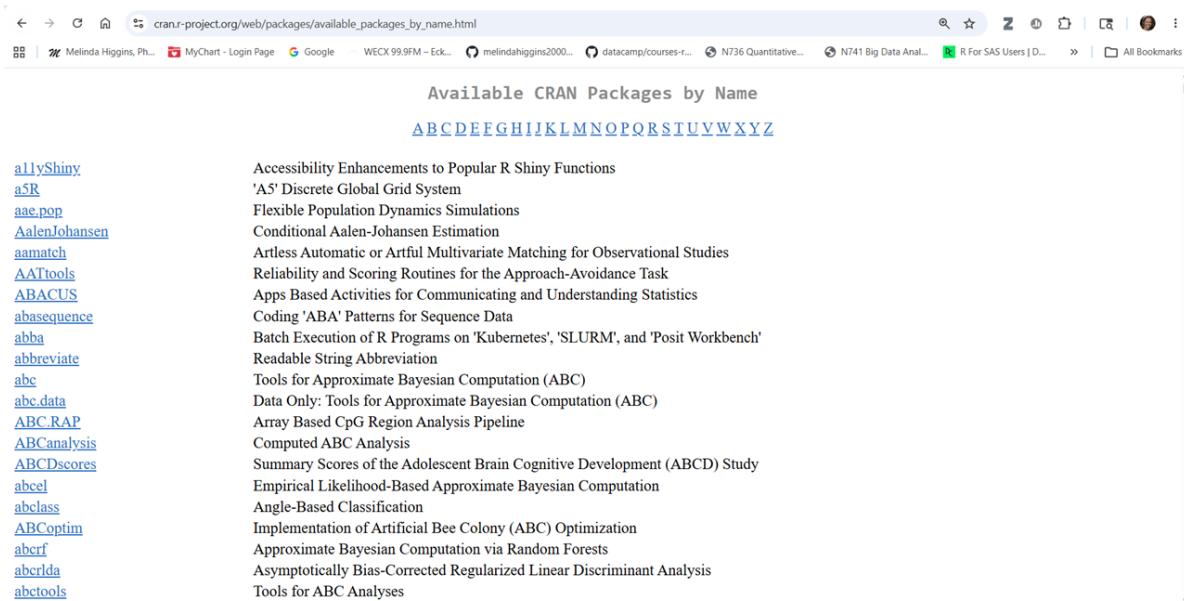
For this course, you will need to have the following packages installed on your computer. All of these packages are available from [CRAN](#).

#### R Packages on CRAN

As of 05/16/2026 at approximately 11:00 am EDT USA time, there are currently 23668 packages hosted on CRAN. More get added everyday!



The screenshot shows the CRAN website interface. On the left is a sidebar with navigation links: CRAN, Mirrors, What's new?, Search, CRAN Team, About R, R Homepage, The R Journal, Software, R Sources, R Binaries, Packages, Task Views, Other, Documentation, Manuals, FAQs, Donations, and Donate. A red arrow points from the 'Packages' link in the sidebar to the 'Available Packages' section on the main page. The main page is titled 'Contributed Packages' and features a box stating 'Available Packages' with the text 'Currently, the CRAN package repository features 23668 available packages.' To the right of this box, in red text, is 'Number of Packages on CRAN as of 05/16/2026; 11:00 EDT USA'. Below the box are links for 'Table of available packages, sorted by date of publication' and 'Table of available packages, sorted by name'. Further down, there is a section for 'CRAN Task Views' and 'Installation of Packages'.



The screenshot shows a web browser window with the URL `cran.r-project.org/web/packages/available_packages_by_name.html`. The page title is "Available CRAN Packages by Name". Below the title is a navigation bar with letters A through Z. The letter 'A' is highlighted. On the left side, there is a list of package names starting with 'A', including `a11yShiny`, `a5R`, `aae.pop`, `AalenJohansen`, `aamatch`, `AATtools`, `ABACUS`, `abasequence`, `abba`, `abbreviate`, `abc`, `abc.data`, `ABC.RAP`, `ABCanalysis`, `ABCDscores`, `abcel`, `abclass`, `ABCOptim`, `abcrf`, `abcrlda`, and `abctools`. On the right side, there is a list of package descriptions corresponding to these names, such as "Accessibility Enhancements to Popular R Shiny Functions" for `a11yShiny` and "Tools for ABC Analyses" for `abctools`.

Available CRAN Packages by Name

[A](#) [B](#) [C](#) [D](#) [E](#) [F](#) [G](#) [H](#) [I](#) [J](#) [K](#) [L](#) [M](#) [N](#) [O](#) [P](#) [Q](#) [R](#) [S](#) [T](#) [U](#) [V](#) [W](#) [X](#) [Y](#) [Z](#)

|                               |                                                                               |
|-------------------------------|-------------------------------------------------------------------------------|
| <a href="#">a11yShiny</a>     | Accessibility Enhancements to Popular R Shiny Functions                       |
| <a href="#">a5R</a>           | 'A5' Discrete Global Grid System                                              |
| <a href="#">aae.pop</a>       | Flexible Population Dynamics Simulations                                      |
| <a href="#">AalenJohansen</a> | Conditional Aalen-Johansen Estimation                                         |
| <a href="#">aamatch</a>       | Artless Automatic or Artful Multivariate Matching for Observational Studies   |
| <a href="#">AATtools</a>      | Reliability and Scoring Routines for the Approach-Avoidance Task              |
| <a href="#">ABACUS</a>        | Apps Based Activities for Communicating and Understanding Statistics          |
| <a href="#">abasequence</a>   | Coding 'ABA' Patterns for Sequence Data                                       |
| <a href="#">abba</a>          | Batch Execution of R Programs on 'Kubernetes', 'SLURM', and 'Posit Workbench' |
| <a href="#">abbreviate</a>    | Readable String Abbreviation                                                  |
| <a href="#">abc</a>           | Tools for Approximate Bayesian Computation (ABC)                              |
| <a href="#">abc.data</a>      | Data Only: Tools for Approximate Bayesian Computation (ABC)                   |
| <a href="#">ABC.RAP</a>       | Array Based CpG Region Analysis Pipeline                                      |
| <a href="#">ABCanalysis</a>   | Computed ABC Analysis                                                         |
| <a href="#">ABCDscores</a>    | Summary Scores of the Adolescent Brain Cognitive Development (ABCD) Study     |
| <a href="#">abcel</a>         | Empirical Likelihood-Based Approximate Bayesian Computation                   |
| <a href="#">abclass</a>       | Angle-Based Classification                                                    |
| <a href="#">ABCOptim</a>      | Implementation of Artificial Bee Colony (ABC) Optimization                    |
| <a href="#">abcrf</a>         | Approximate Bayesian Computation via Random Forests                           |
| <a href="#">abcrlda</a>       | Asymptotically Bias-Corrected Regularized Linear Discriminant Analysis        |
| <a href="#">abctools</a>      | Tools for ABC Analyses                                                        |

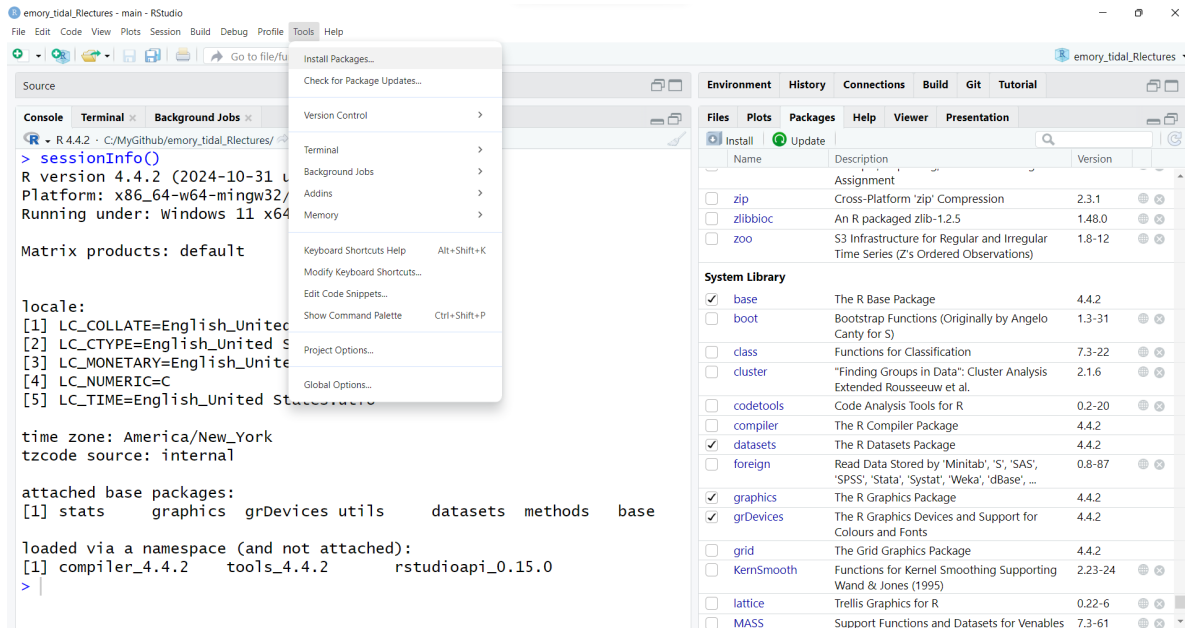
For this course, you will need these 16 packages:

1. `tidyverse` which includes a suite of 9 packages (and a few others as well) that are all installed at the same time:
  - `ggplot2`
  - `dplyr`
  - `tidyr`
  - `readr`
  - `purrr`
  - `tibble`
  - `stringr`
  - `forcats`
  - `lubridate`
  - `haven` - *this package is included in tidyvers*
2. `ggtext`
3. `easystats` which is also a suite of 10 packages:
  - `effectsize`
  - `report`
  - `correlation`
  - `modelbased`
  - `bayestestR`
  - `see`
  - `parameters`
  - `performance`
  - `insight`
  - `datawizard`
4. `gmodels`
5. `gtsummary`
6. `effects`
7. `emmeans`
8. `broom`
9. `rmarkdown`
10. `knitr`
11. `tinytex` - *OPTIONAL* - needed to “knit to PDF”
12. `olsrr`
13. `modelsummary`
14. `janitor`
15. `data.table`
16. `arsenal`

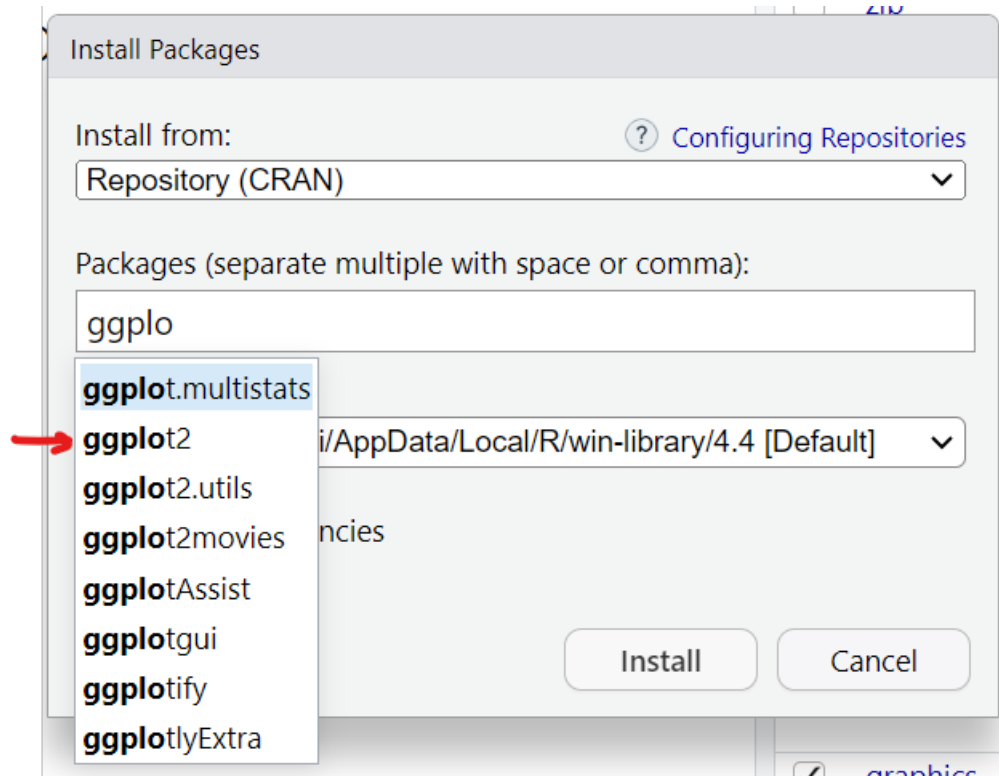
To install these packages, the easiest way is to use the RStudio Menu “Tools/Install” Packages interface:

For example, let’s install a “new” R package, like `ggplot2`.

Go to the RStudio menu “Tools/Install” Packages



This will then open up a window where you can type in the name of the package you want. As soon as we start typing `ggplot2` the menu begins listing all packages with that partial spelling...



You'll notice that there are 3 parts to the installation:

1. Where you want to get the package from (*i.e.*, *which repository - more on repositories later in this course*).
2. The name of the package. You can actually type more than one package name at a time separated by commas if you want to install several packages at once.
3. The file location on your computer where the new package is installed - your file location will be different than mine. But this is useful to know in case something goes wrong. **I would suggest keeping the default settings.**

Install Packages

Install from: **1** [? Configuring Repositories](#)

Package Repository (CRAN) ▼

Packages (separate multiple with space or comma): **2**

ggplot2

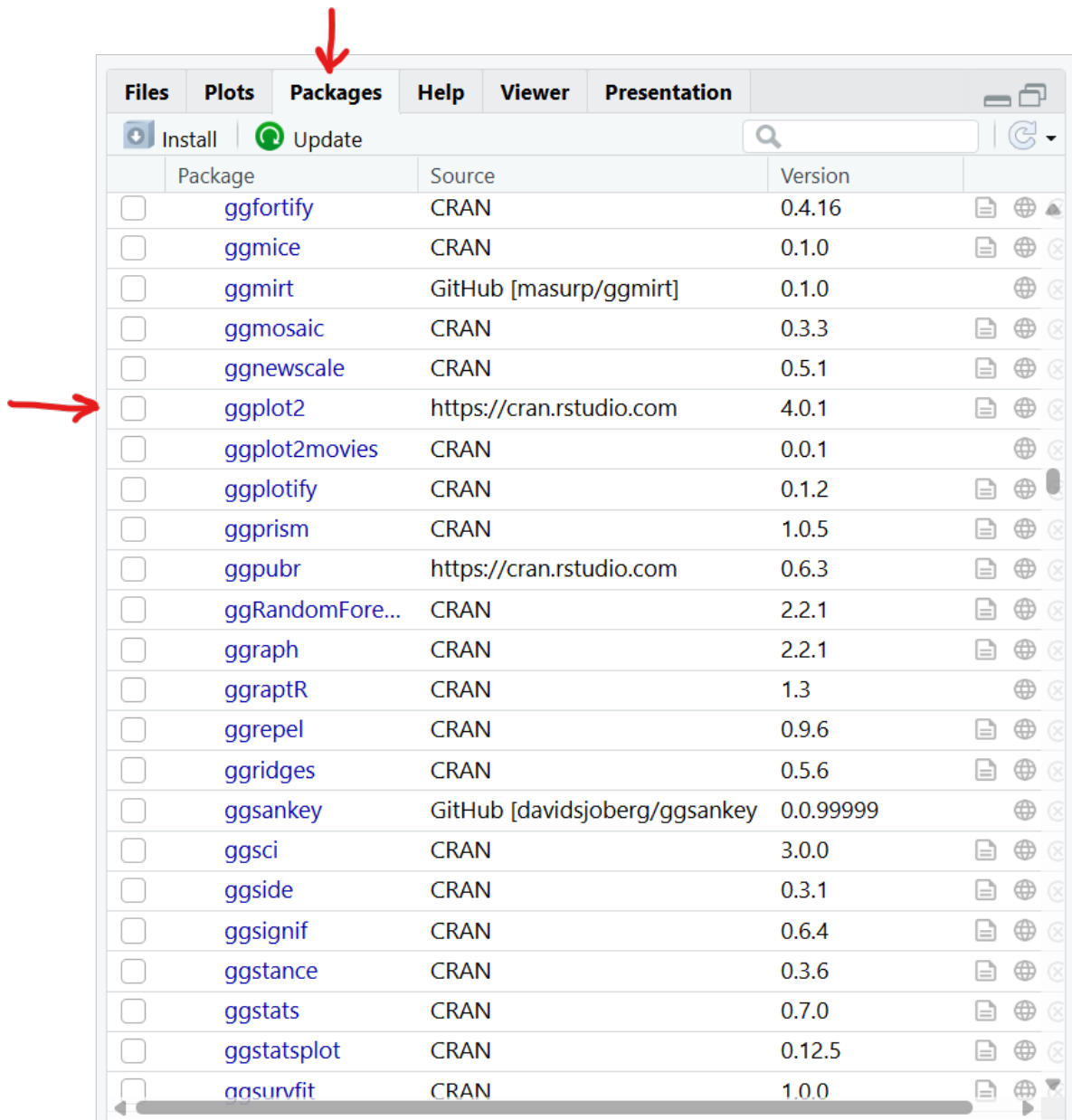
Install to Library: **3**

C:/Users/mkhiggi/AppData/Local/R/win-library/4.6 [Default] ▼

☒ Install dependencies

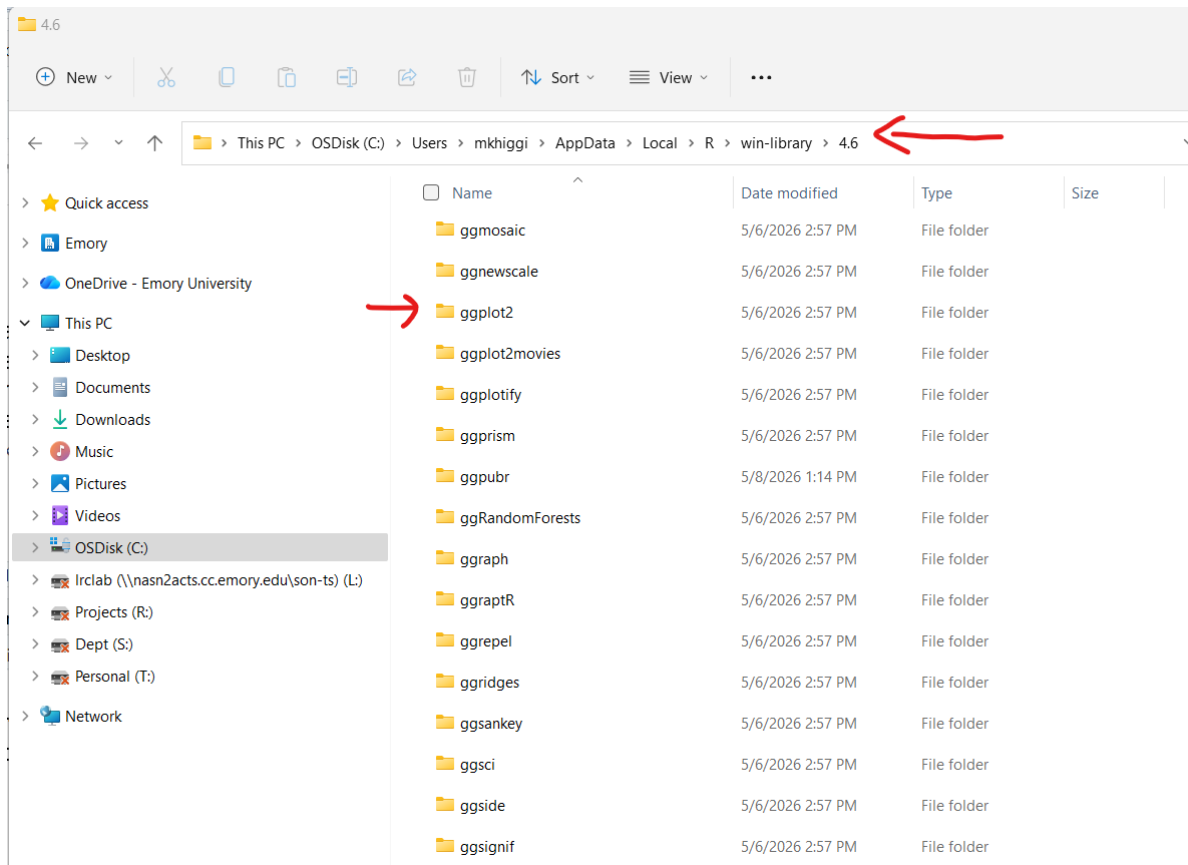
Install Cancel

After installing `ggplot2` you should see `ggplot2` listed under the Packages window:



Repeat this process to install the rest of the R packages needed.

Note: You can also confirm the installation of the R packages on your computer by going to the folder listed under #3 “Install to library” path listed above – on my computer my R packages are in this folder on my computer `C:\Users\mkhiggi\AppData\Local\R\win-library\4.6`.



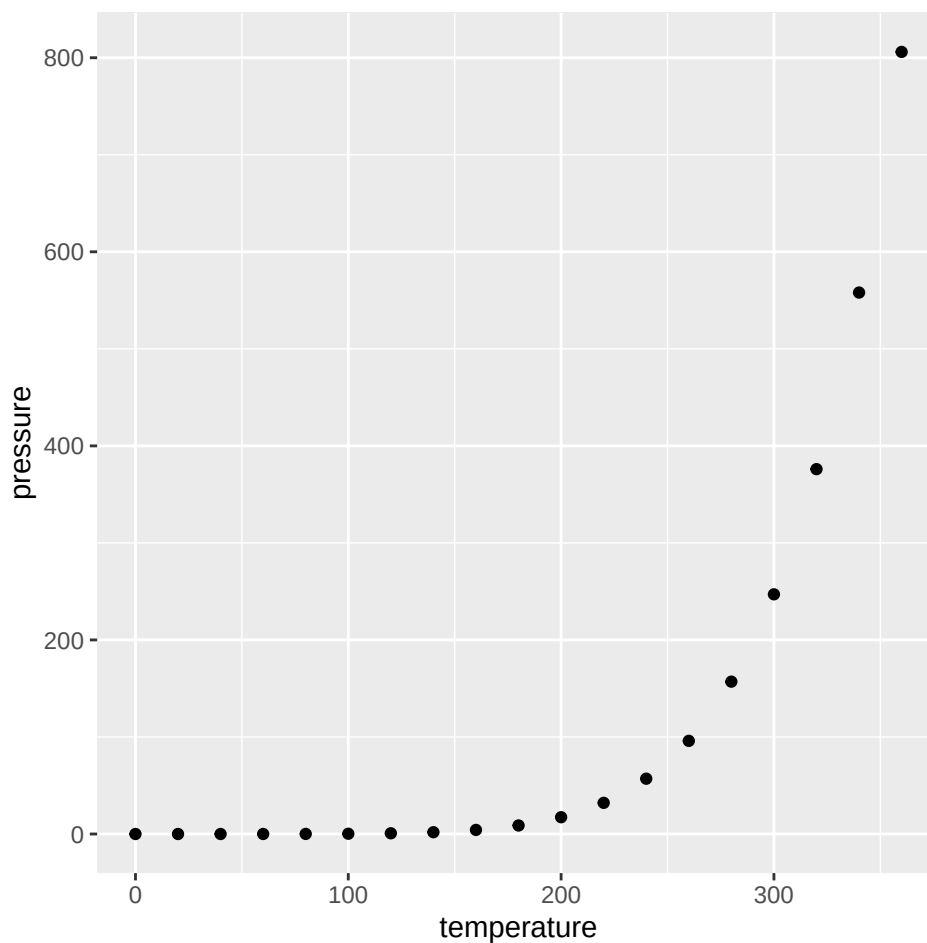
## Learn More About Installing R Packages

- Chapter 1.3 in *Statistical Inference via Data Science: A ModernDive into R and the Tidyverse (Second Edition)*

## 4. Run Quick Check

Let's confirm that your installation of R, RStudio and your packages are working. For example, let's test that your installation of `ggplot2` is working. Run the following R code as a quick test.

```
library(ggplot2)
ggplot(pressure, aes(temperature, pressure)) +
  geom_point()
```



**CONGRATULATIONS** You should be good to go!

## References

- Grolemund, Garrett, and Hadley Wickham. 2011. “Dates and Times Made Easy with lubridate.” *Journal of Statistical Software* 40 (3): 1–25. <https://www.jstatsoft.org/v40/i03/>.
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- Wickham, Hadley, Mara Averick, Jennifer Bryan, et al. 2019. “Welcome to the tidyverse.” *Journal of Open Source Software* 4 (43): 1686. <https://doi.org/10.21105/joss.01686>.
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- Wickham, Hadley, Romain François, Lionel Henry, Kirill Müller, and Davis Vaughan. 2026. *Dplyr: A Grammar of Data Manipulation*. <https://dplyr.tidyverse.org>.
- Wickham, Hadley, and Lionel Henry. 2025. *Purrr: Functional Programming Tools*. <https://purrr.tidyverse.org/>.
- Wickham, Hadley, Jim Hester, and Jennifer Bryan. 2026. *Readr: Read Rectangular Text Data*. <https://readr.tidyverse.org>.
- Wickham, Hadley, Davis Vaughan, and Maximilian Girlich. 2025. *Tidyr: Tidy Messy Data*.

<https://tidyr.tidyverse.org>.

## **Other Helpful Resources**

[Other Helpful Resources](#)